

SIEMENS

POWERMOBIL

SP

Replacement of Parts

Replacing the cable module

© Siemens AG 2003

The reproduction, transmission or use of this document or its contents is not permitted without express written authority. Offenders will be liable for damages. All rights, including rights created by patent grant or registration of a utility model or design, are reserved.

Print No.: SPR2-240.841.01.01.02

Replaces: n.a.

English

Doc. Gen. Date: 05.03

Chapter	Page	Rev.
All	All	01

Document revision level

This document corresponds to the version/revision level effective at the time of system delivery.

Revisions to hardcopy documentation are not automatically distributed.

Please contact your local Siemens office to order current revision levels.

Disclaimer

The installation and service of equipment described herein has to be performed by qualified personnel who are employed by Siemens or one of its affiliates or who are otherwise authorized by Siemens or one of its affiliates to provide such services.

Assemblers and other persons who are not employed by or otherwise directly affiliated with or authorized by Siemens or one of its affiliates are directed to contact one of the local offices of Siemens or one of its affiliates before attempting installation or service procedures.

	Page
1 General	1 - 1
Notes and symbols	1 - 1
Prerequisite	1 - 2
Safety Information	1 - 2
Documents required	1 - 3
Required tools, test equipment and aids	1 - 3
2 Disassembly	2 - 1
Preparations	2 - 1
Covers	2 - 2
X-ray image intensifier and I.I. mini voltage supply	2 - 3
POWERPHOS.	2 - 4
Toothed belt and concealed cable run in the C-arm.	2 - 6
Removing the C-arm and replacing the C-arm cable	2 - 7
Cables within the energy chain	2 - 8
Cable guide in the horizontal carriage.	2 - 9
Removing the cable module and replacing the flange	2 - 10
Removing the horizontal carriage from the vertical tube	2 - 11
3 Installation	3 - 1
Drawing in the cables (horizontal carriage and vertical tube)	3 - 1
Threading in C-arm	3 - 5
Concealed cable guide and toothed belt in the C-arm	3 - 6
POWERPHOS.	3 - 7
I.I. mini voltage supply and X-ray image intensifier	3 - 8
Cables for basic unit.	3 - 9
C-arm cover	3 - 10
Covers of the horizontal carriage	3 - 10
4 Checks and Adjustments	4 - 1
Setting the center position of the cable module to the orbital center position of the C-arm	4 - 1
Setting the sensor of the motorized orbital movement	4 - 3
Preparations	4 - 3
Checking the sensor setting	4 - 3
Setting the sensor	4 - 4
Checking the movements	4 - 4
Checking the camera rotation	4 - 4
Checking the lifting column movement	4 - 5
Checking the orbital movement	4 - 5
Checking the angulation movement	4 - 5
Checking the horizontal movement	4 - 5
Emergency stop of motorized unit movements	4 - 5

	Page
Braking forces / braking moments	4 - 6
POWERPHOS settings.	4 - 6
Covers	4 - 6
Image quality	4 - 6
Ground Wire Test.	4 - 6

Notes and symbols

Emphasized texts in technical documentation have the following meaning:



DANGER indicates an immediate danger that if disregarded will lead to death or serious physical injury.



WARNING indicates a possible danger that if disregarded may lead to death or serious physical injury.



CAUTION used with the safety alert symbol indicates a possible danger that if disregarded will lead to slight or moderate physical injury and/or damage to property.



NOTICE used without the safety alert symbol indicates a possible danger that if disregarded may or will lead to an undesirable result or state other than death, physical injury or property damage.

Prerequisite

NOTE

These instructions apply to the replacement of the cable module in the POWERMOBIL system.

NOTE

2 persons are required to carry out any extensive work.

Estimated time required for 2 persons is 3 days.

We recommend that you replace the cable module in a separate room (workshop).

Please make sure that sufficient working space is available as the basic unit must be accessible from all sides when replacement work is carried out.

Safety Information

 WARNING

Avoiding light to severe, possibly fatal injuries and avoiding material damage.

Product-specific safety information provided in the system manual and the SPR2-240.898.01... service manual as well as the safety information contained in the ARTD, part 2, must be observed.

After all work has been completed and all covers have been attached, perform the protective conductor test according to ARTD-002.731.17. The protective conductor resistance must not exceed 0.2 ohms.

Tests and settings that need to be performed under X-radiation are identified with the radiation symbol . While performing procedures marked with this symbol, radiation protective clothing must be worn.

Documents required

POWERMobil Service Manual	SPR2-240.898.01...
POWERMobil IQ-Quick Test	SPR2-240.820.01...
POWERMobil Wiring Diagram	G5481
POWERMobil - System Manual	SPR2-240.802.01...

Required tools, test equipment and aids

NOTE

All tools, test equipment and aids, with the exception of those marked "*", are listed with specifications in the STC (Service Tool Catalogue).**

- Tool case *
- 1 set of Allen keys * for setting the angulation brake
- 30 mm socket* for disassembling the cable module
- 55 mm socket*
- Ratchet for 30 mm and 55 mm socket*
- Ejector for SUB-D connector pins *
- Ejector for AMP connector pins *
- Bench of approx. 60 cm x 100 cm (if possible, height-adjustable) for disassembling / assembling X-ray image intensifier and for temporary storage of the horizontal carriage
- 200 N spring balance (e.g. 44 15 113 RH090)
- Squared timber * (approx. 60 mm x 70 mm x 100 mm)
- Extractor with 2 clasps* Clasp width approx. 23 cm
- Loctite 221*
- 1 set of resolution tests (e.g. 28 71 820 RE999)
- 1 set of X-ray filters (e.g. 97 98 596 G5321)
- Precision radiation filter 2.1 mm CU (e.g. 99 00 598 XE999)
- 25 mm AL measuring stand, type 26765 acc. to DIN 6868 Part 50* or
- 1.2 mm CU and (e.g. from set 97 98 596 G5321)
- 17 micrometer CU strips (e.g. 11 67 662 G5247)
- Centering cross (e.g. 96 60 051 RE999)
- Dynamic test case (e.g. 37 90 156 X1963 or 97 50 001 X1963)
- Torque wrench 3 Nm - 20 Nm (e.g. 34 24 553)
- Digital multimeter (e.g. "Fluke 8060 A", Part No. 97 02 101)
- Ground wire test meter (e.g. "Safety Tester Bender UNIMET 1000 ST", 51 38 727)
- Service PC running WINDOWS 95 operating system*
- Service Software *
- Serial interface cable "Service PC - SIREMOBIL" (e.g. Part No. 99 00 440)

This page intentionally left blank.

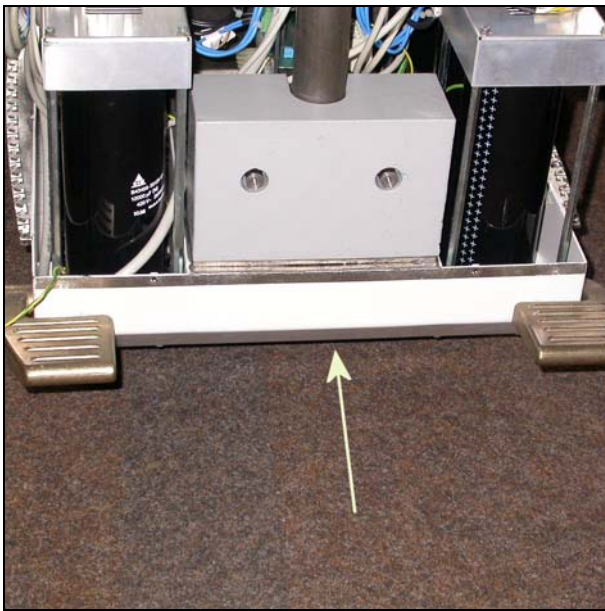


Fig. 1

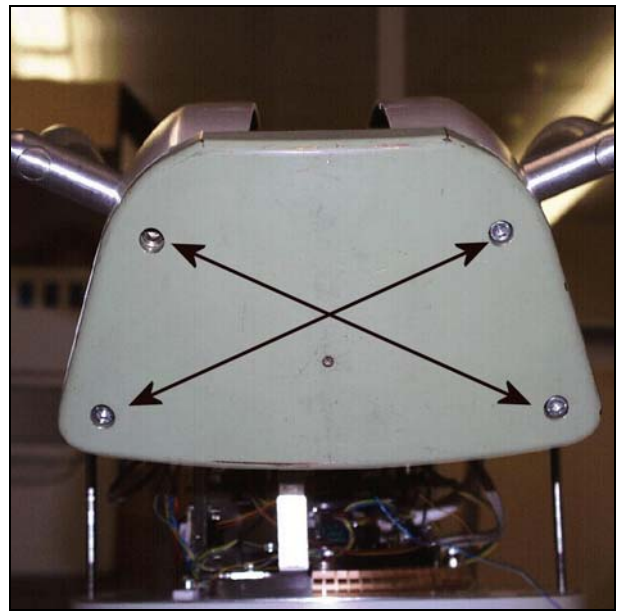


Fig. 2

Preparations

- Position the basic unit in the room so that it can be accessed from all sides.
- Lock the foot brake.
- Push the squared timber underneath the basic unit as indicated by the arrow in Fig. 1 to prevent the basic unit from tilting back after removing the POWERPHOS and the X-ray image intensifier.
- Switch the POWERMOBIL on, and move the lifting column to its approximate vertical center position.
- Switch the POWERMOBIL off again, and unplug the monitor cable X10 plug from the basic unit.
- Unplug the monitor cart's power plug to completely disconnect the monitor cart from the power supply.

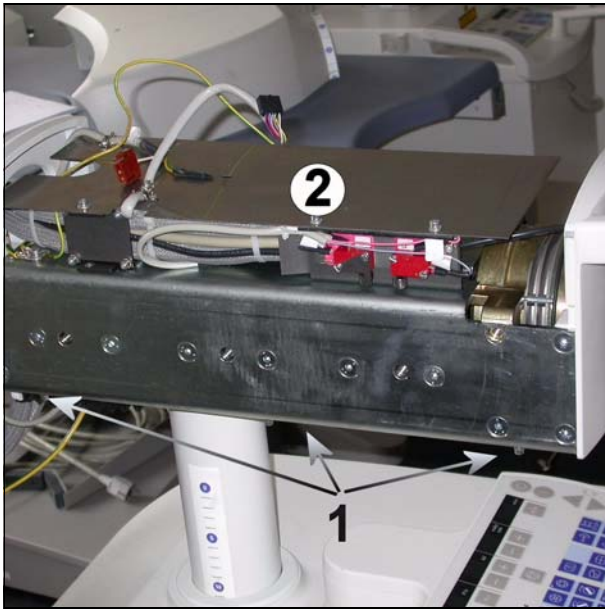


Fig. 3

Covers

⚠ WARNING

Risk of injuries!

If not observed, slight to medium physical injuries and/or material damage can occur.

The C-arm cover contains counterweights and weighs > 10 kg. Firmly hold the C-arm cover when removing.

Before releasing the C-arm cover make sure that the C-arm is in orbital center position.

If POWERPHOS and X-ray image intensifier are still attached to the C-arm, do not move the C-arm to the orbital end positions after loosening the C-arm cover.

- Remove the cover of the basic unit. At the same time disconnect the ground wire connection. (See chapter 1 in the POWERMOBIL service manual.)
- Remove the front C-arm cover from the X-ray image intensifier side of the C-arm (Fig. 2).
- Remove the POWERPHOS cover (lateral paneling screws). (See chapter 1 in the POWERMOBIL service manual.)
- Remove the cover of the horizontal carriage.
 - For this purpose, release 3 screws on each side (1/ Fig. 3).
 - Release the plug-in connections for the brake keypad and the Emergency Stop switch.
 - Remove the protective ground wire from the cover.



Fig. 4



Fig. 5

X-ray image intensifier and I.I. mini voltage supply

- Move the X-ray image intensifier into service position. (See chapter 1 and Fig. 4 in the service manual.)
- Disconnect the I.I. mini voltage supply (high-voltage section and control section). (See chapter 6 in the service manual. Please observe the safety information on discharging the I.I. mini voltage supply.)
- Disconnect the HV cable, as well as E3, E2 and E1 cables.
- Disconnect the CON1 connector.
- Eject the black cathode cable of the X-ray image intensifier (CON1.8)
- Remove the protective ground wire from the X-ray image intensifier and disconnect the I.I. mini voltage supply.
- Disconnect the M4 X1 camera plug from the support bracket, and remove it.
- Disconnect the M30 X1 plug.
- Loosen the cable ties. All image intensifier cables are now disconnected.
- Remove the X-ray image intensifier from the C-arm, and store it in a dark place (avoid exposure to direct sunlight). Avoid exposing the CCD sensor to light and dirt by covering the camera PC board.
- Remove the I.I. mini voltage supply from the C-arm.
- The X-ray image intensifier section of the C-arm contains a round LEMOSA plug connection. Loosen this plug connection (Fig. 5).

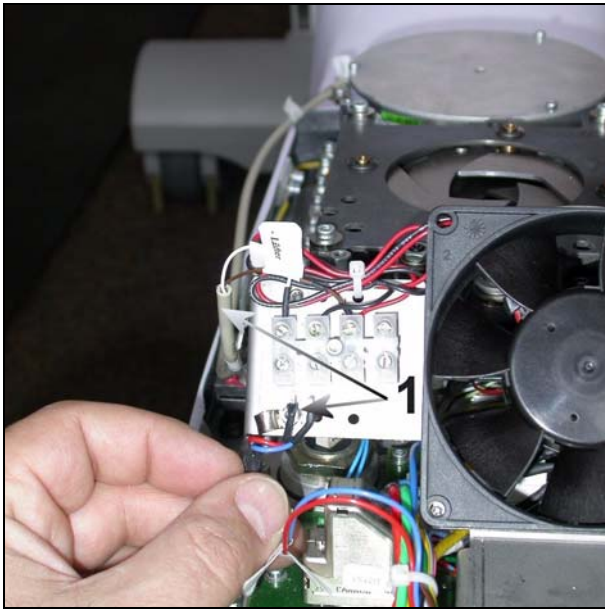


Fig. 6

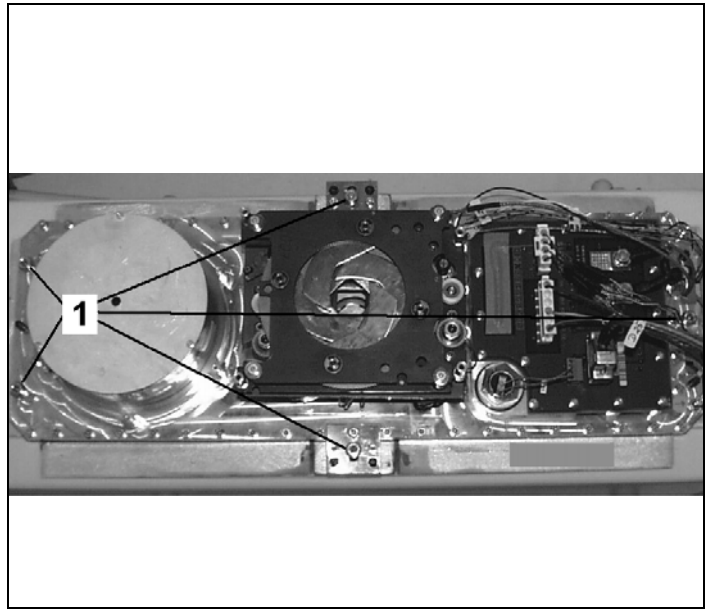


Fig. 7

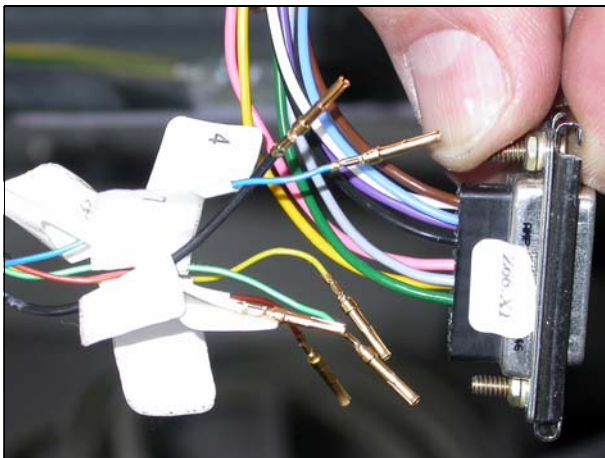


Fig. 8

POWERPHOS

- If applicable, dismantle the DIAMENTOR dose measuring chamber and loosen the connector. Also remove the insulating disk underneath the DIAMENTOR dose measuring chamber.
- Remove the cover above the collimator. The collimator remains attached to the POWERPHOS.
- Remove the fan component from the POWERPHOS.
- Remove the cable ties of the fan cable.
- Note down the connection order of the fan assembly supply cable, and loosen the cables (terminals and colors of the cables, see 1/ Fig. 6).
- Also loosen the cable for the fan integrated in the C-arm. Also note down the connection order (1/ Fig. 6).

- Loosen the POWERPHOS mounting screws (1/Fig. 7). Do not loosen any other screws!
- Loosen all POWERPHOS plug connections.
- Loosen the two inverter connections (terminal screws).
- Loosen the POWERPHOS protective ground wire.
- Loosen the two clamps on the front of the POWERPHOS.

NOTE

To facilitate lifting the POWERPHOS out of the C-arm, thread 2 cable ties (min. 60 x 6 mm) into the existing holes.

- Lift the POWERPHOS out of the C-arm and set it down.
- Eject all labeled contact pins from the Z66 X1 plug (Fig. 8).
- Remove any existing cable ties.

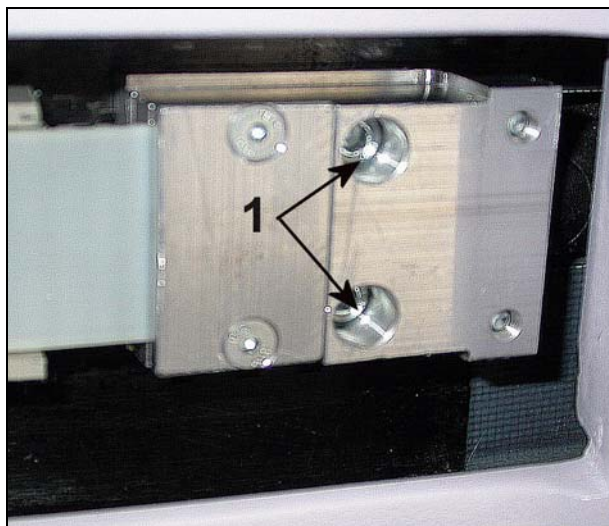


Fig. 9

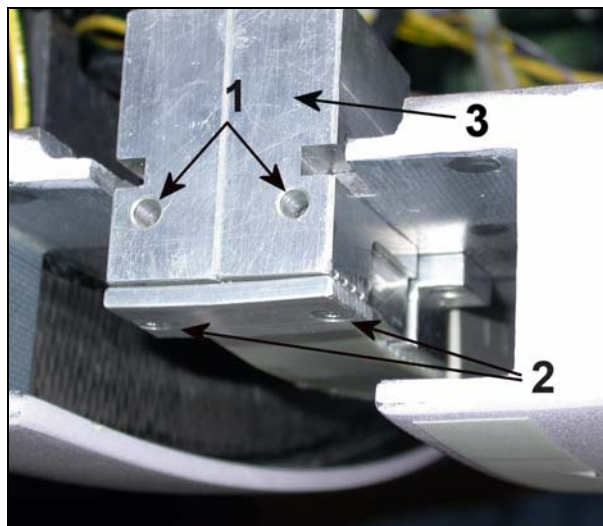


Fig. 10

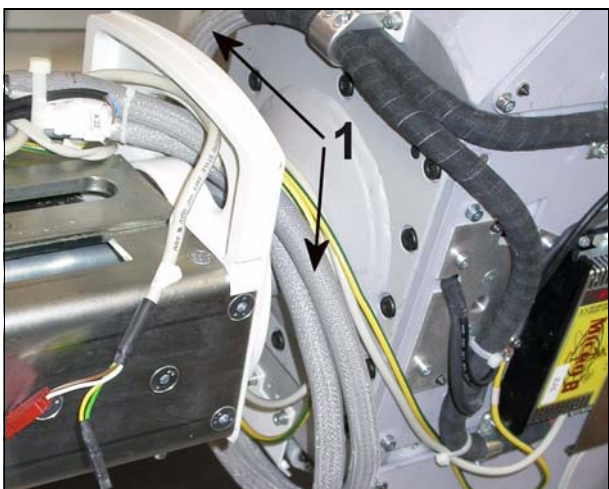


Fig. 11

Toothed belt and concealed cable run in the C-arm

- The C-arm position is in 0° orbital (center position) and 180° angulation (X-ray image intensifier below, tube on top, one turn of the cable around the angulation axis, see 1/ Fig. 11).
- Release the two tension screws on the image intensifier side (1/ Fig. 10); the tension on the toothed belt is released.
- Unscrew the two mounting screws (2/ Fig. 10) of the clamping plate. The toothed belt is released.
- Loosen the two screws (1/ Fig. 10), and remove the clamping piece (3/ Fig. 10).
- Pull the C-arm cable out of the C-arm.
- Release the two tube-side screws (1/ Fig. 9), and pull the C-arm cable out of the C-arm.

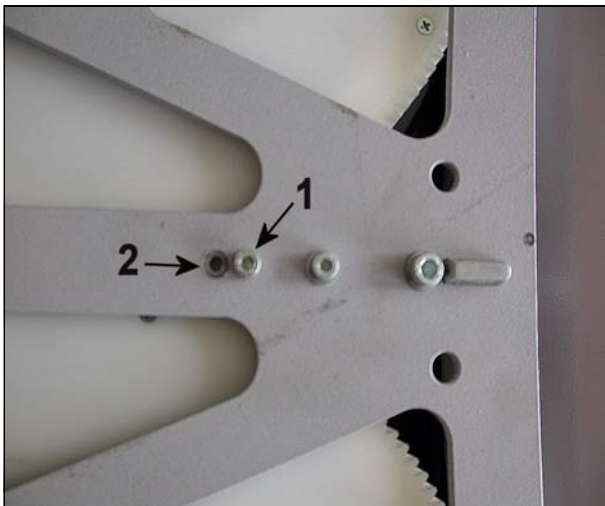


Fig. 12



Fig. 13



Fig. 14

Removing the C-arm and replacing the C-arm cable

- Block the cable module. For this purpose, move the screw (1/Fig. 12) from the thread hole (2/Fig. 12) to the thread hole (1/Fig. 12) and tighten it.
- Remove the holders of the covers of the cable modules on both sides (1/Fig. 13).
- Have 2 persons turn the C-arm out of the bearing, as shown in Fig. 14. For this purpose, try to take the weight off the C-arm to avoid excess pressure to the C-arm from the last rollers.
- Set the C-arm down.
- The cables in the interior of the C-arm are included in the scope of supply of the cable module. Replace them one-to-one. When pulling out the old cables draw in a pull wire.

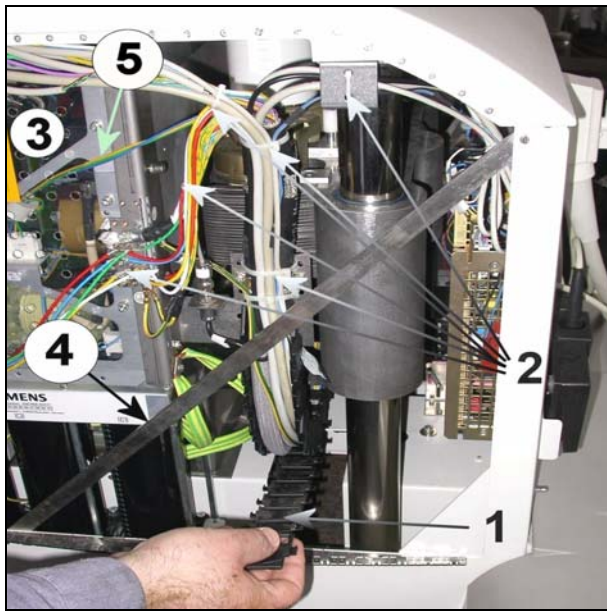


Fig. 15

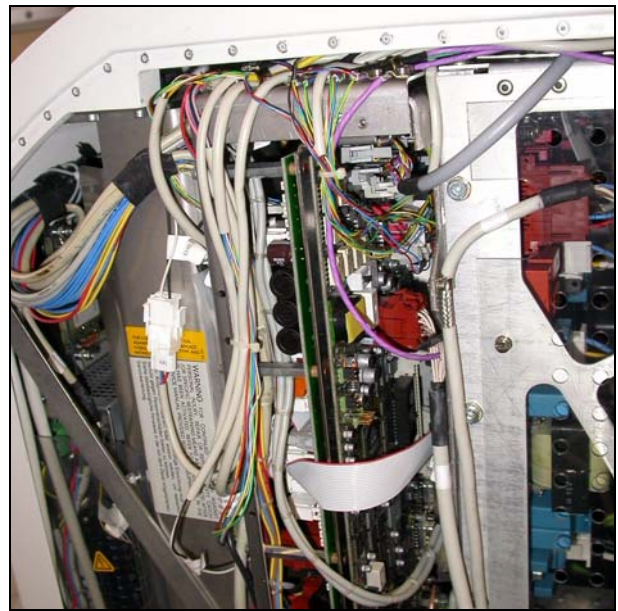


Fig. 16

Cables within the energy chain

- Remove the plate (4/Fig. 15).
- Remove the fabric tape from both ends of the energy chain.
- Open the energy chain fully (1/Fig.15).
- Remove the Plexiglas cover (3/Fig. 15).
- Release the frame of the PC boards D1 and D30 on the opposite side of the basic unit, and pull it out to the side (Fig. 16).
- Remove the ferrite core from the M2 X1 cable (5/Fig. 15).
- Trace the cables inside the energy chain, and loosen the corresponding cable ties, cable clamps and plug/screw connections in the basic unit.
- Pull any cables back towards the energy chain to facilitate threading the cables in and out.

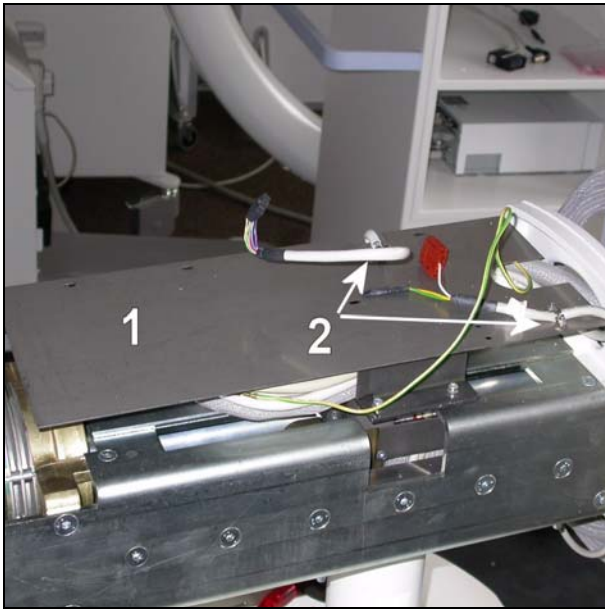


Fig. 17

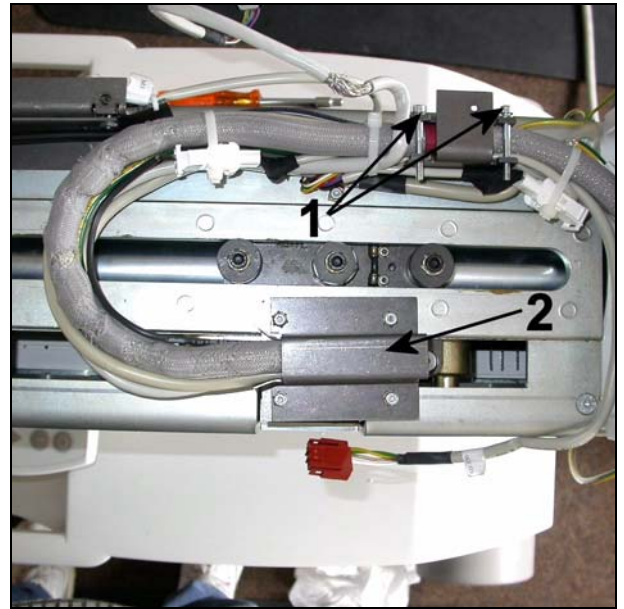


Fig. 18

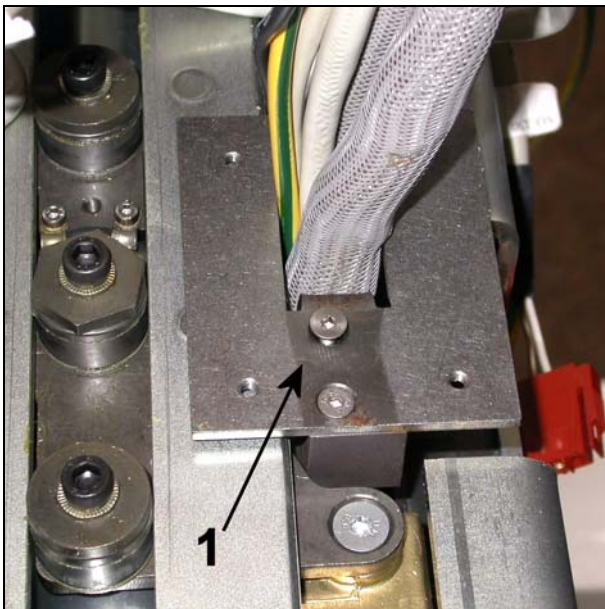


Fig. 19

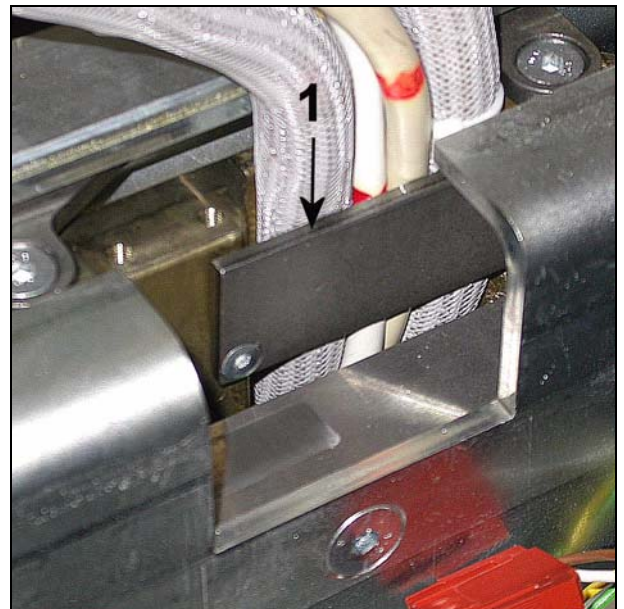


Fig. 20

Cable guide in the horizontal carriage

- Remove the cover above the cable (1/Fig. 17). Also loosen the cable clamps (2/Fig. 17).
- Unscrew the two cable guides, see (1/Fig. 18) and (2/Fig. 18).
- Unscrew the cable guide (1/Fig. 19).
- Unscrew the cable guide (1/Fig. 20). After moving the horizontal carriage, the second screw of the cable guide will be accessible.

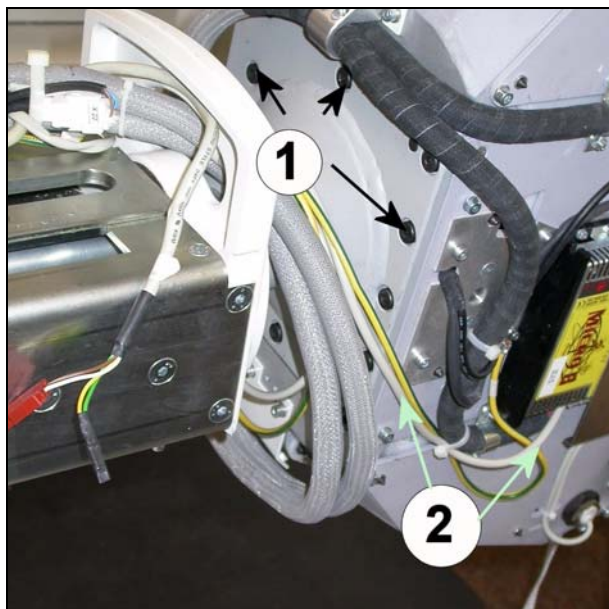


Fig. 21

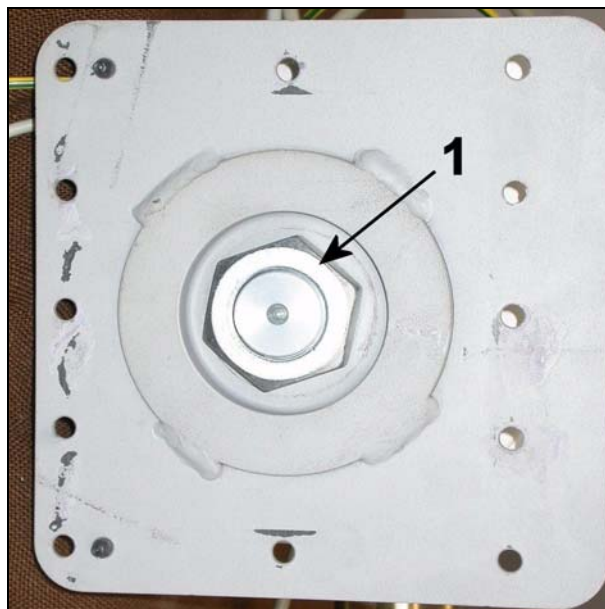


Fig. 22

Removing the cable module and replacing the flange

NOTE

The cable module is attached to the flange with 12 screws. When removing the cable module, one person needs to prevent it from falling. The cables of the module are not yet free. For this reason, you should set the cable module down on a suitable base close to the horizontal carriage.

- Loosen the cable module from the flange (12 screws (1/Fig. 21)).
- Set the cable module down on a suitable base close to the horizontal carriage.
- Loosen the ground wire connection on the cable module and the ST1 connector of the cable module electronic unit (2/Fig. 21). Cut open the corresponding cable ties.
- Loosen the nut (1/Fig. 22) using a 55 mm socket and a ratchet. Remove nut and lock washer.
- Use an extractor to remove the flange from the axle of the horizontal carriage.
- Loosen the new flange from the new cable module (12 screws (Fig. 21)).
- Take the new lock washer from the scope of supply of the cable module. Discard the old lock washer.
- Place the new flange onto the axle of the horizontal carriage, and fasten it using the new lock washer and the nut (1/Fig. 22). Do not change the angulation during this procedure, and avoid rotating the angulation shaft.



Fig. 23



Fig. 24

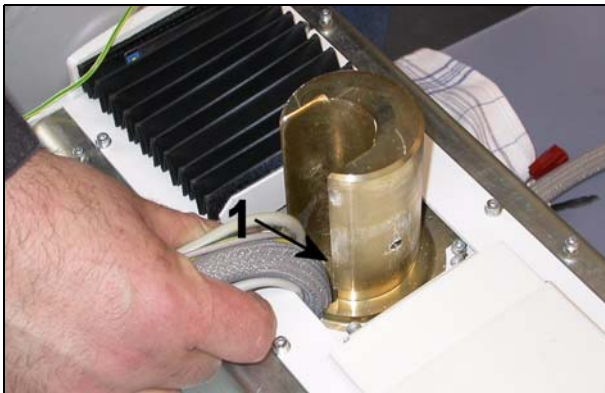


Fig. 25



Fig. 26

Removing the horizontal carriage from the vertical tube

NOTE

To facilitate removal of the cables remove the horizontal carriage. The cable with the M3 X30 plug remains in the vertical tube of the lifting column and in the horizontal carriage.

- Unscrew the two countersunk screws from their positions (1/ Fig. 23).
- Lift the horizontal carriage out from the tube, and set it down on a suitable base. Make sure that you pull out the cables in the vertical tube only far enough to be able to set down the horizontal carriage.
- Remove the cables from the vertical tube.

NOTE

The D30 X3 plug will not fit through the guide channel (1/ Fig. 25) of the horizontal carriage.

- Eject the contacts of the D30.X3 plug, and note them down.
- Remove the cables from the guide channel (1/ Fig. 25). Start with the smallest plug. For this purpose, position the horizontal carriage vertically and hold it.

This page intentionally left blank.

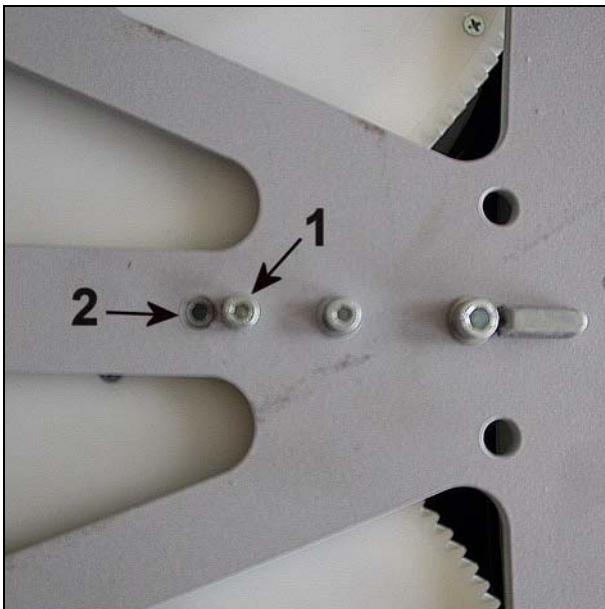


Fig. 1



Fig. 2

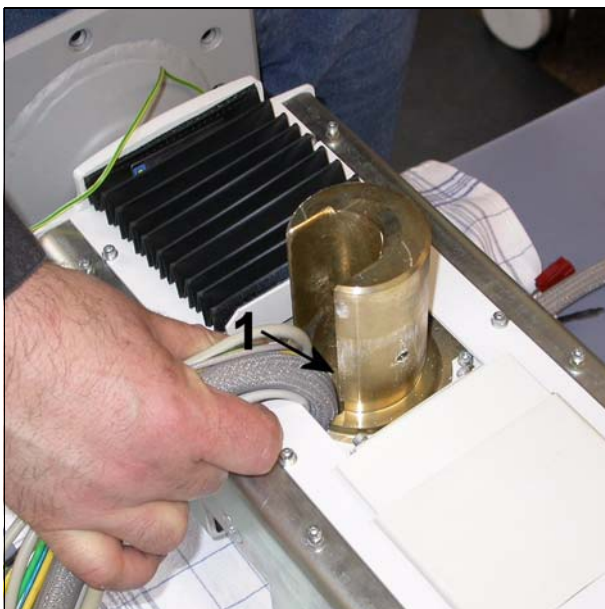


Fig. 3

Drawing in the cables (horizontal carriage and vertical tube)

- Instead of the old cable module place the new one close to the horizontal carriage.
- Lock the cable drums of the new cable module. For this purpose, move the locking screw (1/Fig. 1) from position (2/Fig. 1) to position (1/Fig. 1) and screw it in.

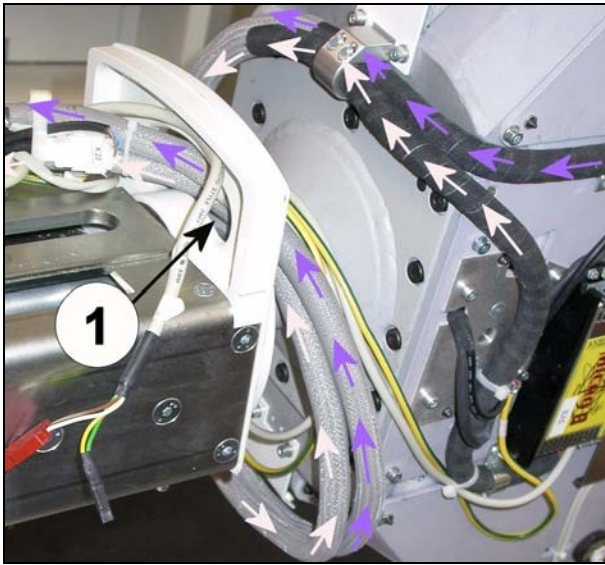


Fig. 4

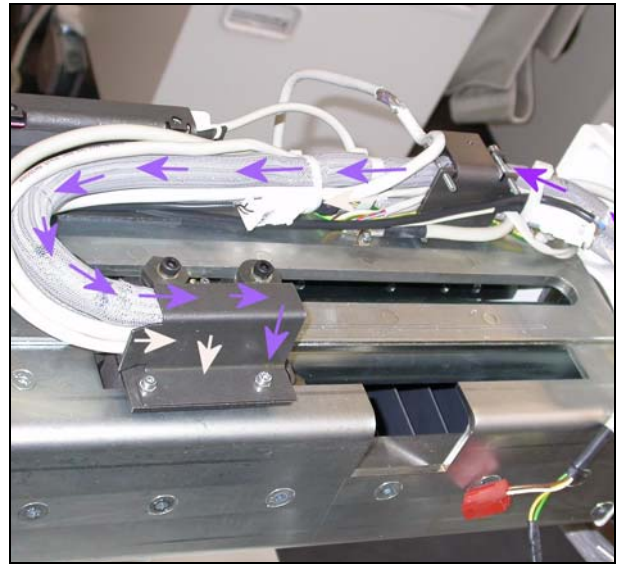


Fig. 5

- Locate the D30 X3 plug on the new cable module, and eject the leads. At the same time, note down the pin assignment.

NOTE

Fig. 4 and Fig. 5 show the position of the two cables of the cable module after assembly. When you run the cables and draw them into the horizontal carriage, make sure that the cables do not intersect at any point. To facilitate drawing the cables into the horizontal carriage guide (Fig. 3) start with the largest plug.

- Pull both cables through the recess in the horizontal carriage (1/ Fig. 4). Make sure that the cables do not intersect after assembly.
- Pull all leads and plugs of the two cables of the cable module through the horizontal carriage guide. Make sure that the cables do not intersect. The cables drawn in previously should now be positioned between the two thick cables of the cable module. You should still be able to move the cables once they have been pulled cleanly into the guide.
- Draw the cables into the vertical tube using a guide wire. Make sure that the cables inside the vertical tube do not intersect.
- When all cables have been pulled in, refit the horizontal carriage on the vertical tube. Make sure that no cables are crushed.
- Refit the horizontal carriage using the two countersunk screws.
- Attach the cable module to the flange so that the cables will settle with one turn around the angulation axis. Make sure that the cables do not intersect (Fig. 4).
- Reattach the cable module using the 12 screws. Apply Loctite 221 to the screws.

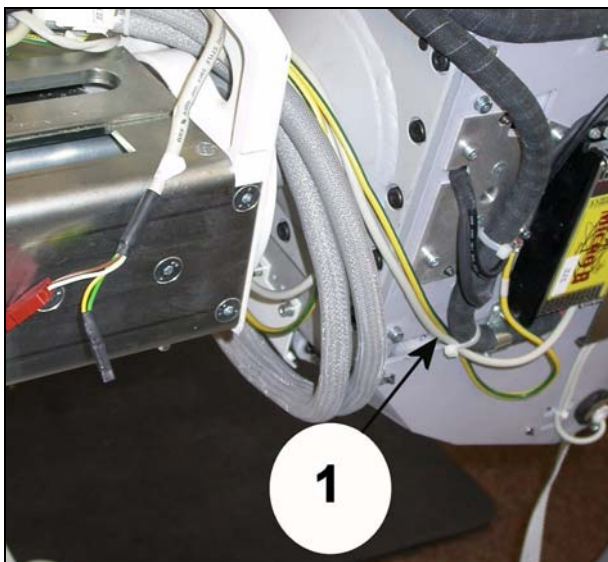


Fig. 6

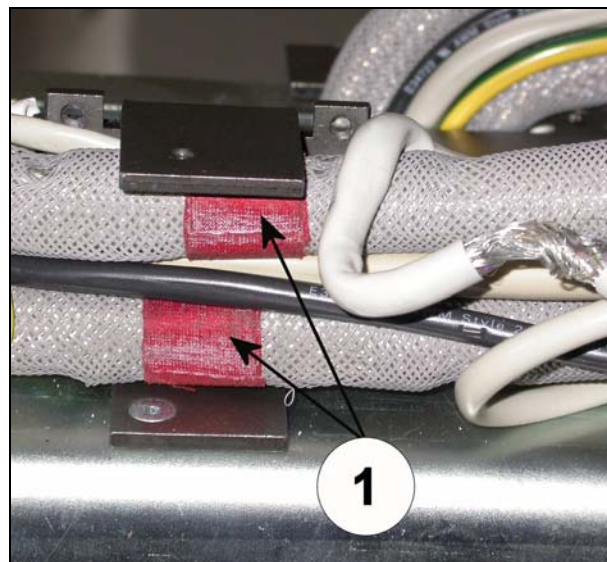


Fig. 7

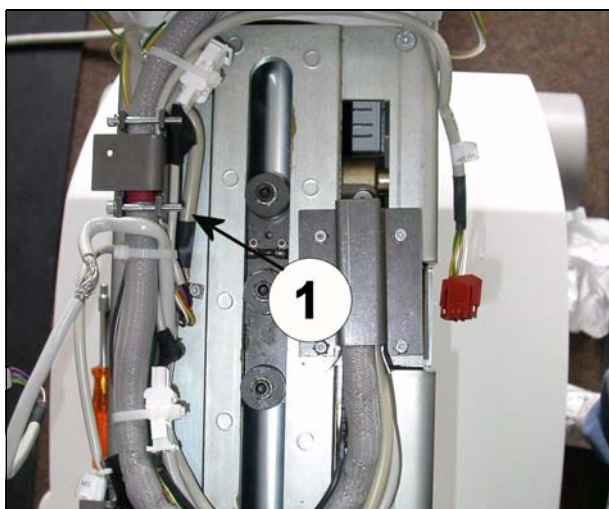


Fig. 8

- Run and connect the protective ground wire and the cable (1/Fig. 6) to the new cable module. These cables also lie with one turn around the angulation axis.
- Fasten the two cables to the cable module using cable ties (Fig. 6).

NOTE

Run the cables in the horizontal carriage from the cable module towards the guide, and attach them.

- The two thick cables for the cable module have a red mark running round them (1/Fig. 7). Insert and secure these cables together with the other cables of the cable module in the clamping point (1/Fig. 8).
- To check the cable lengths rotate the cable module angulation from end position to end position. The cable loop around the angulation axis must be sufficiently long, but with only slight excess length.

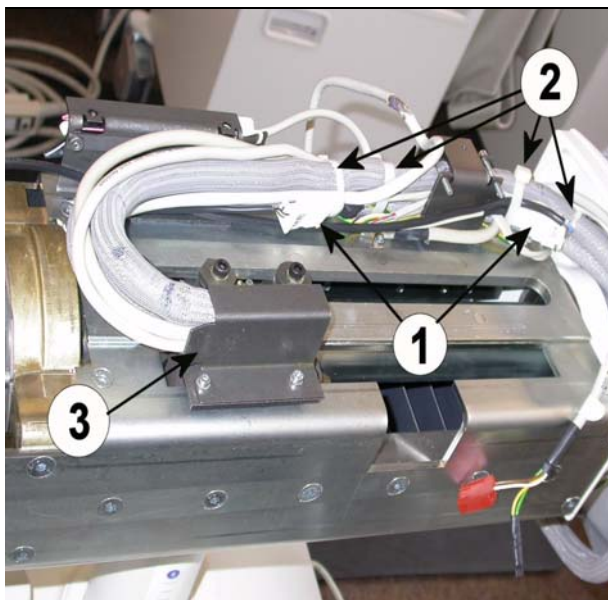


Fig. 9

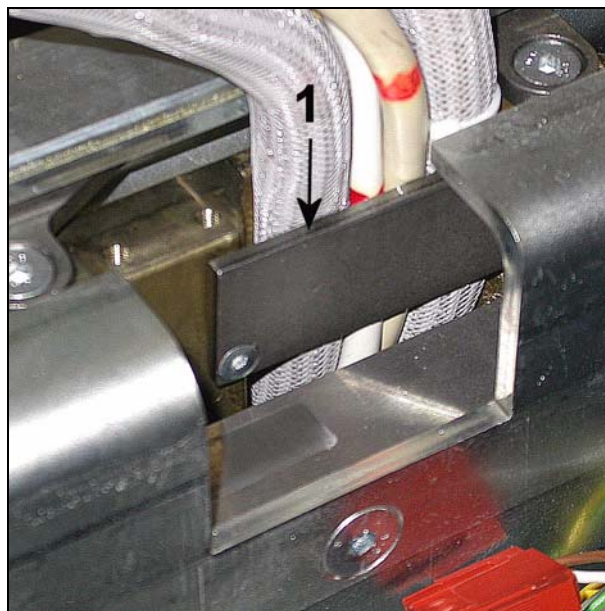


Fig. 10

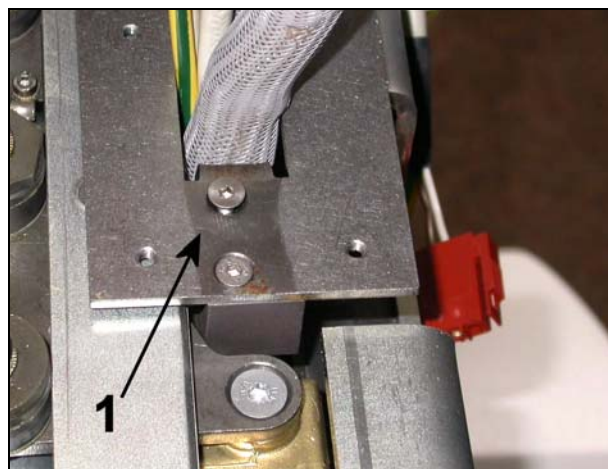


Fig. 11

- Fig. 9 shows the horizontal carriage in its end position (moved out fully). Adapt the cable lengths in the horizontal carriage as illustrated in Fig. 9. For this purpose, push any excess lengths down into the vertical tube. Some cables have a red mark running round them. They lie in the cable clamp (1/Fig. 10).
- Reconnect the plug connections (1/Fig. 9).
- Reattach cables and plugs using cable ties (2/Fig. 9).
- Refit the cable guide (1/Fig. 11) and cable clamps (1/Fig. 10) and (3/Fig. 9).
- Move the horizontal carriage from end position to end position several times. At the same time, check that the cables run neatly and the loops forming in the end positions are neither too short nor too long. If necessary, adapt cable lengths once again.



Fig. 12

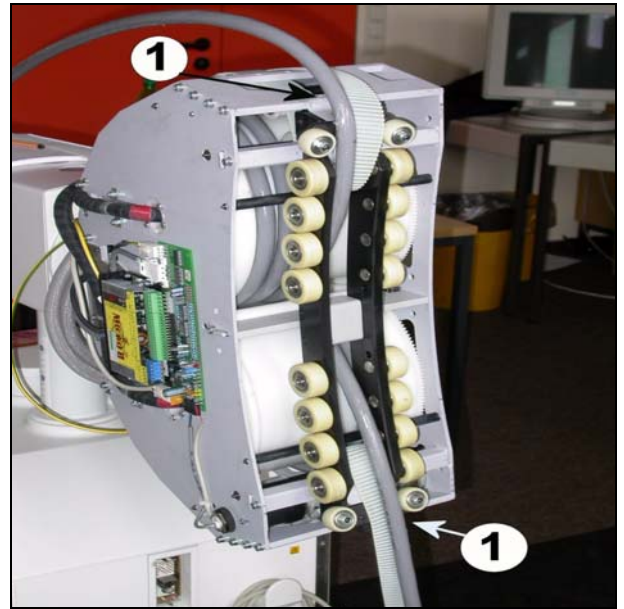


Fig. 13



Fig. 14

Threading in C-arm

- To facilitate threading in the C-arm unscrew the holders for the covers of the cable module present on both sides (1/ Fig. 12).
- The cable module is still blocked (Fig. 1).
- There are 2 rubber buffers attached to the cable module (1/ Fig. 13). If the new cable module does not include these buffers, remove them from the old cable module and attach them to the new module.
- Lift the C-arm with 2 persons, and turn it into the bearing support (Fig. 14). To avoid high pressure on the bearing rollers and the C-arm make sure that the C-arm is not inserted canted.
- Refit the holders on both sides (1/ Fig. 12).

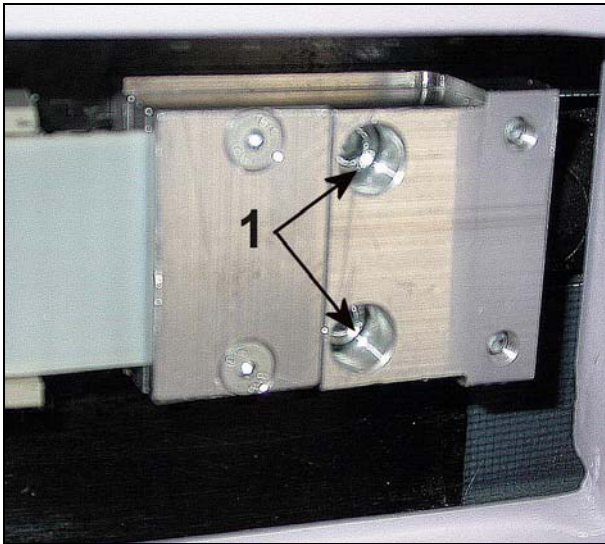


Fig. 15

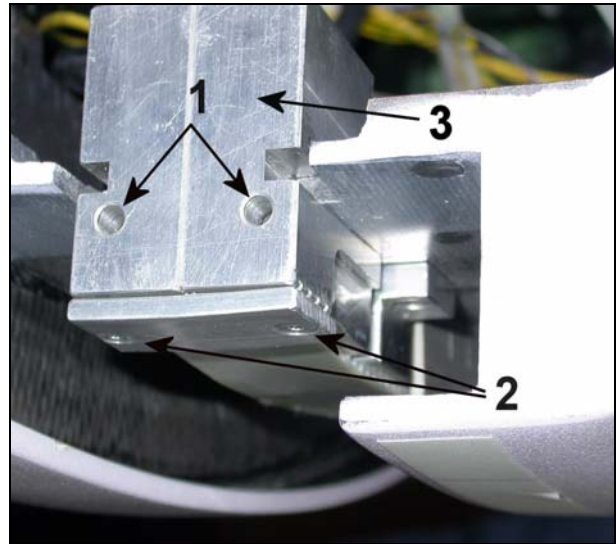


Fig. 16

Concealed cable guide and toothed belt in the C-arm

- Insert the tube-side cable (including toothed belt) in the C-arm's concealed cable guide in the direction of the POWERPHOS. Make sure that the cable is not twisted.
- Draw the cable end with plugs into the POWERPHOS C-arm segment.
- Attach the clamping block (Fig. 15) to the C-arm using the screws (1/Fig. 15).
- Loosen the two mounting screws (2/Fig. 16) of the clamping plate from the cable on the image intensifier side. The toothed belt is released.
- Insert the cable in the C-arm's concealed cable guide, and draw the cable end with plugs into the C-arm at the front.
- Insert the clamping piece (3/Fig. 16) in the C-arm.
- Fasten the toothed belt with the clamping plate and the screws (2/Fig. 16). The belt's teeth are positioned in the slots of the clamping piece.
- Release the cable module block. For this purpose, change the screw (1/Fig. 1) from position (1/Fig. 1) to position (2/Fig. 1) and tighten it.

NOTE

If high tension prevents you from inserting the toothed belt up to the last slots of the clamping piece, position the toothed belt one or more slots closer and attach it there. Move the C-arm approx. $\pm 45^\circ$ out from the orbital center position, but do not move it to the end position of the image intensifier. Then loosen the toothed belt again, increase its tension manually as much as possible, and reattach it.

- Increase the toothed belt tension using the tensioning screws (1/Fig. 16). From the orbital center position move the C-arm $\pm 45^\circ$ several times to eliminate any play of the toothed belt. If necessary, increase the tension of the toothed belt in the slots of the clamping piece, and reattach it.

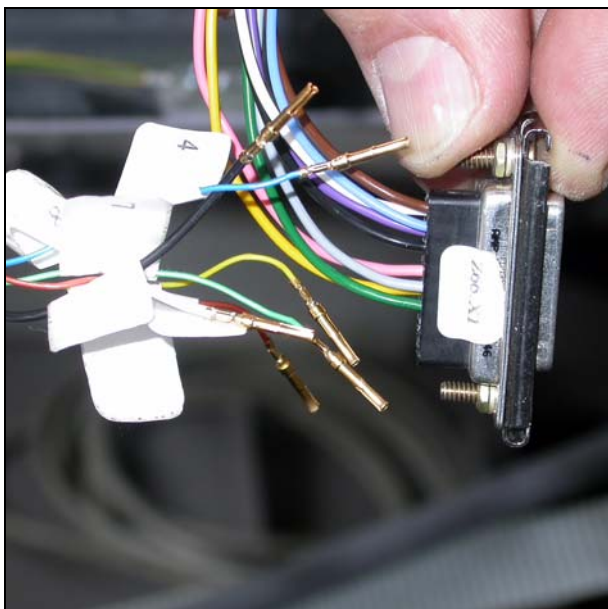


Fig. 17

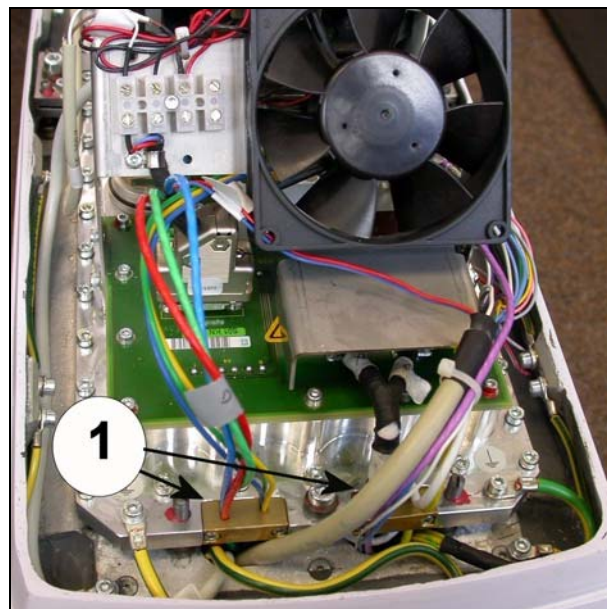


Fig. 18

POWERPHOS

- Run the cables in the POWERPHOS C-arm segment so that the cables are not crushed when inserting the POWERPHOS.
- Run the loose contacts to the plug Z66 X1, and reinsert them (Fig. 17). Tie the cables using cable ties.
- Reinsert the POWERPHOS in the C-arm segment. Make sure that the cables are not crushed.
- Screw the two inverter connectors back on. Tighten the nuts with a torque of 4.8 Nm.
- Fasten the protective ground wires.
- Connect all plugs to the POWERPHOS.
- Connect the collimator plug.
- Attach the sterile cover above the collimator.
- If applicable, assemble the DIAMENTOR dose measuring chamber with insulating disk and attach it.
- Attach the two cable clamps (1/ Fig. 18). Make sure that the cable shields makes good ground contact.
- Fit the fan assembly, and connect it observing the connection order as previously noted.
- Tie or attach all cables using cable ties.
- Fasten the POWERPHOS to the C-arm. Make sure that the screws are tightened evenly.



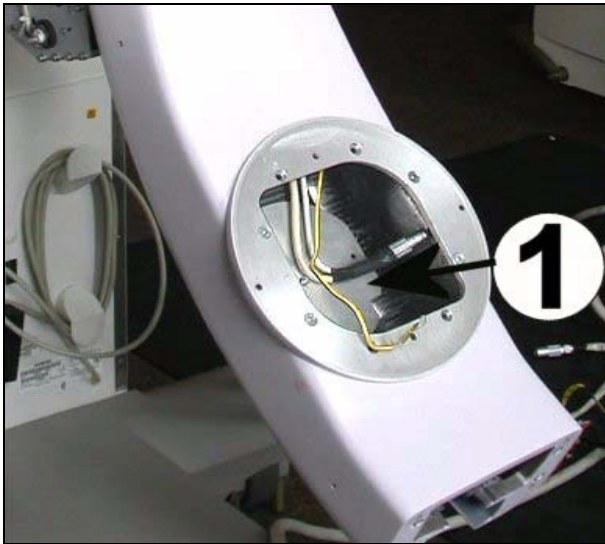


Fig. 19

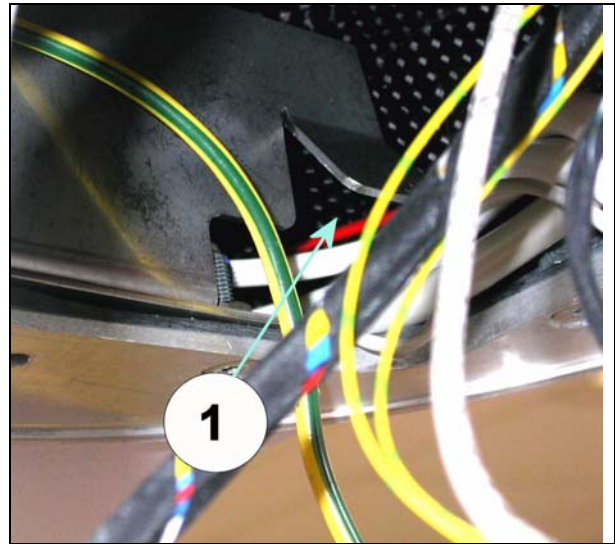


Fig. 20

I.I. mini voltage supply and X-ray image intensifier

NOTE

To avoid blocking camera rotation after fastening the X-ray image intensifier make sure that all cables are run carefully during installation.

- Connect the Lemos plug connector in the C-arm. Insert the plug connector into the C-arm. Run the cable in the C-arm so that it is positioned at the side underneath the metal cover (1/ Fig. 19).
- Insert the control and high-voltage sections of the I.I. mini voltage supply in the C-arm.
- Plug in the connection cable (ribbon cable) between control section and high-voltage section.
- Attach the X-ray image intensifier in the (suspended) service position.
- Fasten the ground wire connections to the image intensifier.
- Connect the high-voltage cable as well as the cables E1, E2 and E3.
- Connect the black cathode cable of the X-ray image intensifier to the CON1.8 plug.
- Connect the CON1 plug to the control section.
- Attach the camera plug M4 X1 to the mounting bracket.
- Connect the plug M30 X1.
- Fasten all cables using cable ties.
- Lift the X-ray image intensifier, and carefully thread in the cables. Guide the high-voltage cable, the cable E3 and the protective ground wire through the bracket (1/ Fig. 20). Insert the ribbon cable between control section and HV section of the I.I. mini voltage supply, at the side between the metal cover (1/ Fig. 19) and the C-arm.
- Fasten the image intensifier.

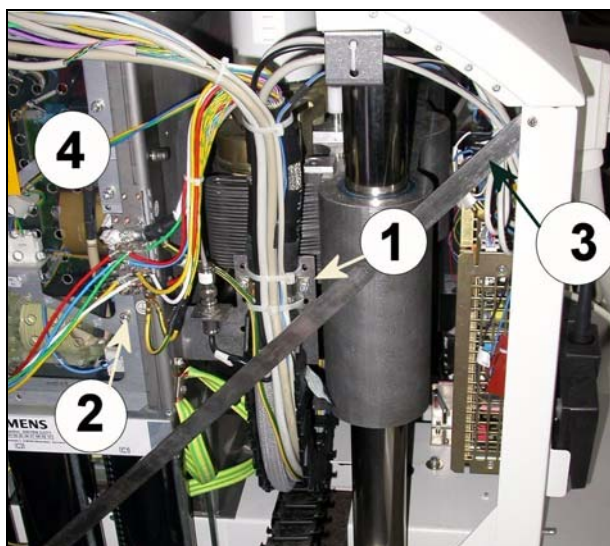


Fig. 21



Fig. 22

Cables for basic unit

- Insert the cables in the energy chain. Make sure that cables do not intersect.
- Fasten both cables for the cable module using cable clamps (1/Fig. 21).
- Close the energy chain.
- Secure the last link on both ends of the energy chain using cable ties or insulation tape to avoid accidental opening.
- Attach the ferrite core to the cable with the plug M2. X1 (1 turn around the ferrite core). Connect the plug.
- Attach the connection cable for the inverter. Tighten the nuts with a torque of 4.8 Nm. Fasten the cable using a cable clamp. Make sure that the shielding makes good ground contact.
- Fasten the protective ground wires.
- Attach the Plexiglas cover (4/Fig. 21).
- Reconnect the BNC plug for the coaxial cable.
- Insert the previously ejected contacts of the plug D30 X3 in the connector housing, as previously noted.
- Run the cables with their plugs to the corresponding plug connections / PC boards according to labeling. Fix the cables using cable ties.
- Tie the cables close to the lifting column using cable ties in such a way that they do not rub against any moving parts or are crushed when moving the lifting column.
- Fix all cables to the cable holder (1/Fig. 21) using cable ties.
- Reattach all shielded cables using the provided cable clamps. Make sure that the shielding makes good contact (2/Fig. 21) (1/Fig. 22).



- When running cables along the frame for the PC boards D1 and D30 (Fig. 22), make sure that they are not crushed when inserting and tightening the frame.
- Fix the run cables using cable ties.
- Refit the plate (3/ Fig. 21).

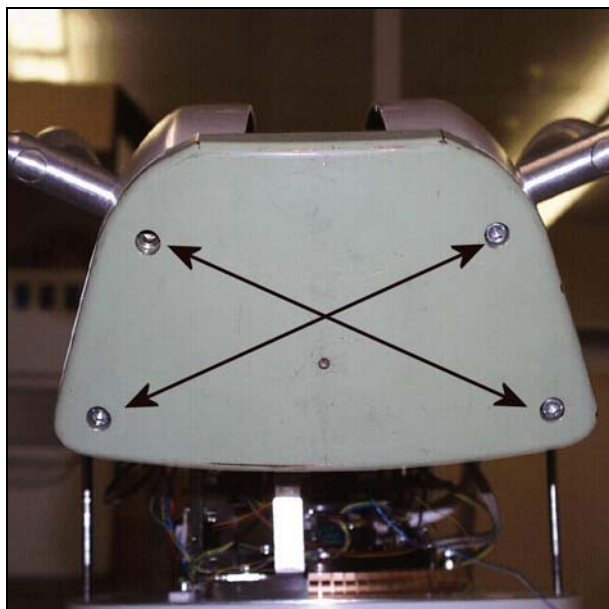


Fig. 23

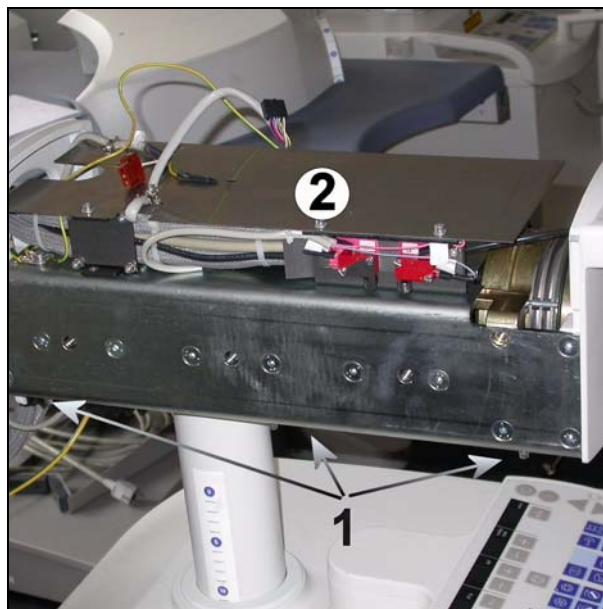


Fig. 24

C-arm cover

- Secure the front C-arm cover to the image intensifier side of the C-arm (Fig. 23). When inserting cables make sure that they are not crushed or torn off.

Covers of the horizontal carriage

- Refit the cover plate (2/ Fig. 24).
- Reattach the cable in the cable clamps. Make sure that the shielding makes good ground contact.
- Screw the protective ground wire to the cover of the horizontal carriage.
- Reconnect the plugs of the brake button and of the Emergency Stop switch.
- Lift the cover above the horizontal carriage and fasten it with screws (1/ Fig. 24).

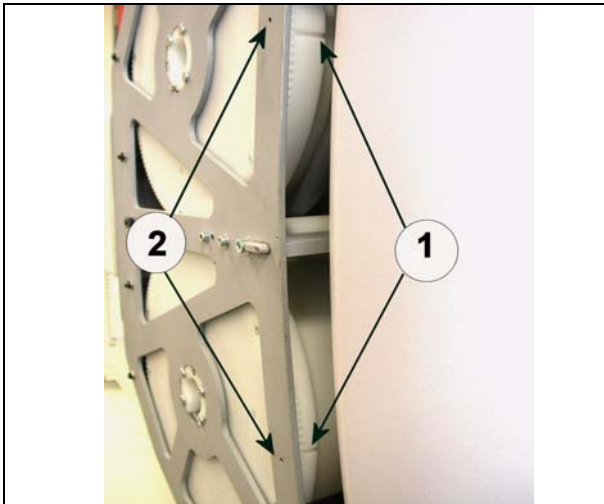


Fig. 1

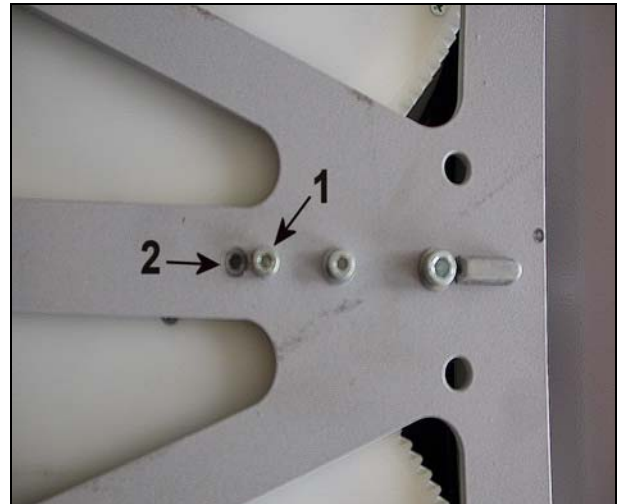


Fig. 2

Setting the center position of the cable module to the orbital center position of the C-arm

- The lock of the cable module is released, the screw (1/Fig. 2) is in position (2/Fig. 2). The C-arm can be moved orbitally.
- Turn the C-arm orbitally so that the markings (1/Fig. 1) are centered on the threaded holes (2/Fig. 1). The cable drums are in center position.
- Lock the cable module in this position. For this purpose, turn the screw (1/Fig. 2) into position (1/Fig. 2).



Fig. 3

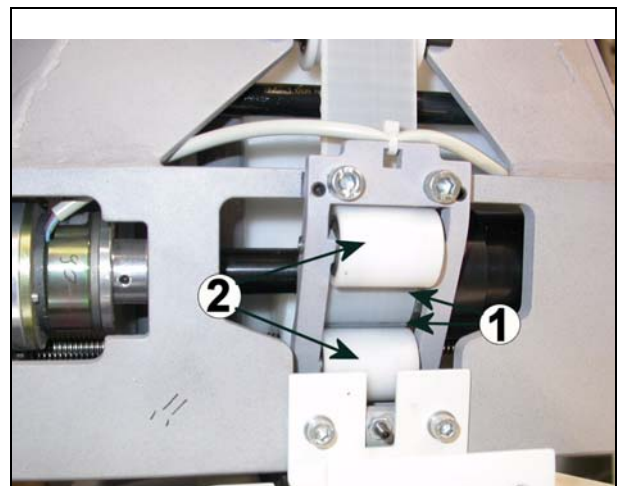


Fig. 4

- Loosen the screw (1/Fig. 3), but do not screw it out. Press the screw (1/Fig. 3) in direction of the arrow (Fig. 3). With this the cable drums are uncoupled from the toothed belt of the C-arm. The C-arm can be moved slightly orbitally.
- Move the C-arm orbitally so that both markings (1/Fig. 4) on the toothed belt are positioned between the two rollers (2/Fig. 4). The C-arm is now in orbital 0° position.

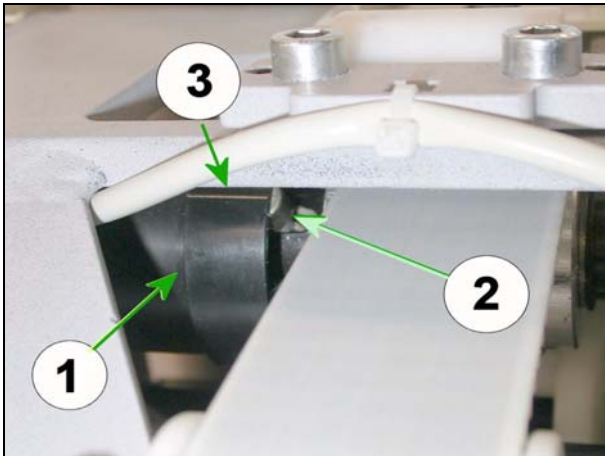


Fig. 5

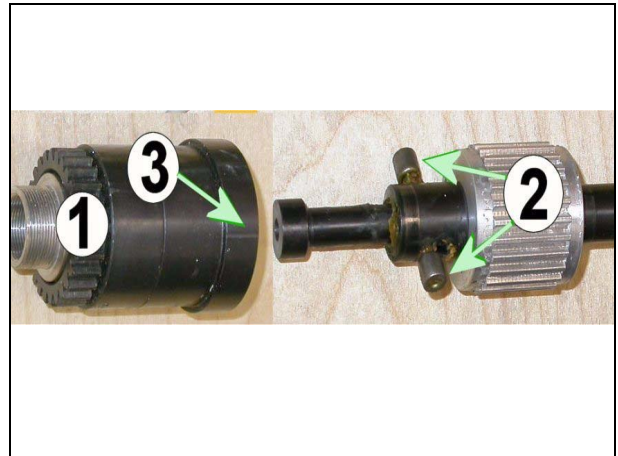


Fig. 6

- Keep the C-arm in this position.
- Turn the bell (1/Fig. 5) until the rollers (2/Fig. 5) are aligned with the marking on the bell (3/Fig. 5). Fig. 6 shows the bell (1/Fig. 6), the marking (3/Fig. 6) and the shaft with the two rollers (2/Fig. 6) once again in a removed state.
- In this position of the bell and rollers, push the pinion gear back into the original position. Make sure that the teeth of the gear engage.
- Retighten the screw (1/Fig. 3).

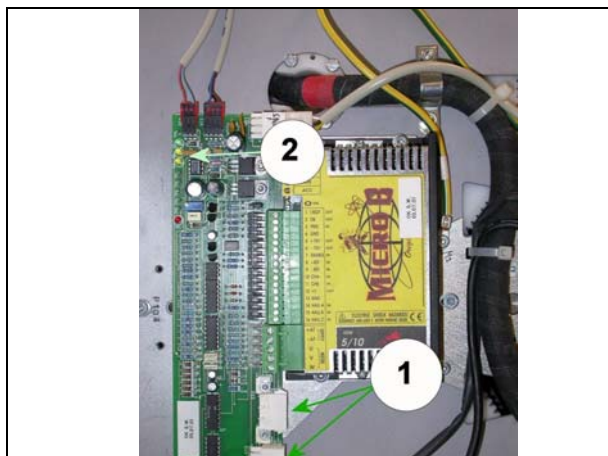


Fig. 7

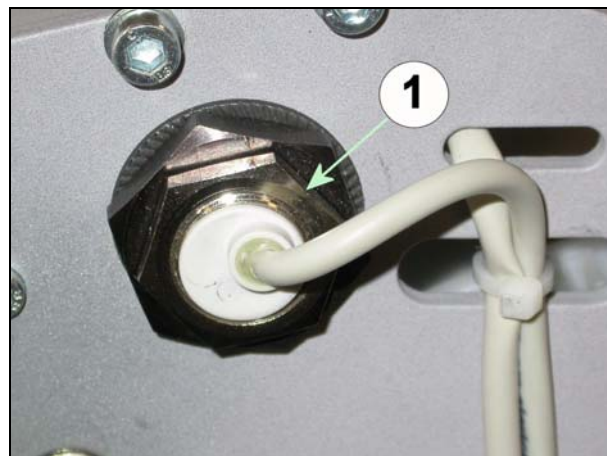


Fig. 8

Setting the sensor of the motorized orbital movement

NOTE

The sensor is already adjusted upon delivery of the cable module. Generally, the setting only needs to be checked according to the section "Checking the sensor setting".

However, should the sensor not be adjusted properly, it can be adjusted according to the "Setting the sensor" section.

Preparations

- Make sure again that all cables, plugs, cable clamps and protective ground wires are connected.
- Connect the basic unit with the monitor cart (plug X10).
- Disconnect both plugs ST5 and ST6 (1/ Fig. 7) from the cable module electronics.
- Connect the monitor cart to the power supply.
- Switch the system on.
- The cable drum is still locked. The screw (1/ Fig. 2) is still screwed into position (1/ Fig. 2).

Checking the sensor setting

- Loosen the orbital brake by pressing the associated button.
- The C-arm can be moved slightly orbitally. At the same time the rollers (2/ Fig. 5) move into the end position inside the bell (1/ Fig. 5). In both end positions the LED D3 (2/ Fig. 7) should light up half-brightly.
- If the check is not successful, go to the next section "Setting the sensor".
- Switch the system off.
- Change the position of the stop screw of the cable module back from position (1/ Fig. 2) to position (2/ Fig. 2) and tighten it.

- Reconnect the plugs ST5 and ST6 on the cable module electronics.
- Switch the system on and check the orbital movement again with released brake.

Setting the sensor

NOTE

**If the previous checking of the sensor setting was successful, this section can be omitted.
Continue with "Checking the movements".**

- The stop screw of the cable module is still in position (1/Fig. 2).
- Loosen the counternut on the sensor (1/Fig. 8). The sensor can now be adjusted by turning it. Loosen any present cable ties.
- Connect a DMM to test point TP2 (-) and TP4 (+) of the cable module electronics (DC mode).
- Loosen the orbital brake by pressing the associated button.
- The C-arm can be moved slightly orbitally. At the same time the rollers (2/ Fig. 5) move into the end position inside the bell (1/Fig. 5).
Set $1.03 \text{ V} \pm 10 \%$ in both end positions by turning the sensor. Turn several times from end position to end position in order to average out the mechanical play. If no DMM is available, the setting can be carried out with the LED D3. The LED D3 may light up only half brightly in the end positions.
- Retighten the counternut. Make sure that the setting of the sensor is not changed again during the process.
- Switch the system off once again.
- If the cable of the sensor is very twisted, disconnect it briefly, untangle it and replace previously removed cable ties.
- Change the position of the cable module's locking screw from position (1/Fig. 1) back to position (2/Fig. 1) and tighten it.
- Reconnect the plugs T5 and ST6 to the cable module electronics.
- Switch the system on, and check the orbital movement again with released brake.

Checking the movements

Checking the camera rotation

- The system is switched on.
- Rotate the camera under fluoroscopy from 0° to $+220^\circ$ and then to -220° degrees.
At the same time, make sure that the camera rotates smoothly up to the end positions.



⇒ There must be no error messages on the display during this process.

Checking the lifting column movement

- The system is switched on.
- Extend and retract the lifting column completely. At the same time, make sure that the energy chain moves properly with it and there are no cables in the way of the mechanically moving parts and make sure cables are not being crushed.
 - ⇒ The lifting column movement must stop on its own in its end positions.

Checking the orbital movement

- Release the orbital brake.
- Turn the C-arm into both orbital end positions.
 - ⇒ The cables of the cable module must be properly wound up and down on the respective cable drum.
The C-arm must not move on its own accord in any position.
- Turn the C-arm orbitally back into 0° position.
- Reactivate the orbital brake.

Checking the angulation movement

- Release the angulation brake.
- Turn the C-arm into both angulation end positions.
 - ⇒ The cables around the angulation axis must move correctly.
 - ⇒ The cables must not be taut in the end positions of the angulation axis.
- Turn the angulation back to the 0° position.
- Reactivate the angulation brake.

Checking the horizontal movement

- Release the horizontal movement brake.
- Move the horizontal carriage from end position to end position.
 - ⇒ The force required should be approximately the same over the entire distance.
- Move the horizontal carriage into several positions and release it.
 - ⇒ The horizontal carriage must not move on its own accord.
- Reactivate the horizontal movement brake.

Emergency stop of motorized unit movements

- Press the Emergency Stop switch
 - ⇒ The vertical lifting column and the motorized support of the orbital movement must not operate any longer.
- Unlock the Emergency Stop switch again.

Braking forces / braking moments

- Check the braking forces/braking moments according to the service instructions (chapter 7, Braking forces / braking moments).

POWERPHOS settings

- Set the POWERPHOS and collimator according to the system service instructions (chapter 6, "POWERPHOS").
- Afterwards, refit the POWERPHOS cover.

Covers

- Refit all covers. Make sure the protective ground wires make good contact.

Image quality

Perform the IQ quick test.

Ground Wire Test

- Perform the protective ground wire test according to ARTD-002.731.17.
 - ⇒ The protective round wire resistance must not exceed 0.2 ohms.